

SQL DATA ANALYSIS PROJECT

Project 3 – SQL Data Analysis

Name: JAISHA N

Internship: DecodeLabs Data Analytics Internship

Project: SQL Data Analysis

Date: June 2026

1. OBJECTIVE

The objective of this project is to analyze an e-commerce dataset using SQL queries and extract meaningful business insights. The project focuses on applying SQL concepts such as SELECT, COUNT, SUM, AVG, GROUP BY, and ORDER BY to perform data analysis.

2. TOOLS USED

- DB Browser for SQLite
- SQLite
- Microsoft Excel
- GitHub

3. DATASET OVERVIEW

The dataset contains order-related information including:

- Order ID
- Date
- Customer ID
- Product
- Quantity
- Total Price
- Payment Method
- Order Status
- Referral Source
- Shipping Details

4. SQL QUERIES AND RESULTS

Query 1: Display Dataset Records

SQL Query:

```
SELECT * FROM "Dataset for Data Analytics (2)" LIMIT 6;
```

Purpose:

To view the dataset and understand its structure.

Result:


```
SELECT SUM(TotalPrice) AS TotalRevenue
FROM "Dataset for Data Analytics (2)";
```

Purpose:

To calculate the total revenue generated.

Result:

SQL 1*	
1	SELECT SUM(TotalPrice) AS TotalRevenue
2	FROM "Dataset for Data Analytics (2)";

TotalRevenue	
1	1264761.96

Query 4: Calculate Average Order Value

SQL Query:

```
SELECT AVG(TotalPrice) AS AverageOrderValue
FROM "Dataset for Data Analytics (2)";
```

Purpose:

To determine the average value of customer orders.

Result:

SQL 1*	
1	SELECT AVG(TotalPrice) AS AverageOrderValue
2	FROM "Dataset for Data Analytics (2)";

AverageOrderValue	
1	1053.9683

Query 5: Analyze Orders by Payment Method

SQL Query:

```
SELECT PaymentMethod, COUNT(*) AS Orders
FROM "Dataset for Data Analytics (2)"
GROUP BY PaymentMethod;
```

Purpose:

To identify the most frequently used payment methods.

Result:

```
SQL 1*  
1 SELECT PaymentMethod, COUNT(*) AS Orders  
2 FROM "Dataset for Data Analytics (2)"  
3 GROUP BY PaymentMethod;  
4  
  
PaymentMethod  Orders  
1 Cash          246  
2 Credit Card   234  
3 Debit Card     232  
4 Gift Card      230  
5 Online        258
```

Query 6: Analyze Orders by Status

SQL Query:

```
SELECT OrderStatus, COUNT(*) AS Orders
FROM "Dataset for Data Analytics (2)"
GROUP BY OrderStatus;
```

Purpose:

To analyze the distribution of order statuses.

Result:

```
SQL 1* x
1 SELECT OrderStatus, COUNT(*) AS Orders
2 FROM "Dataset for Data Analytics (2)"
3 GROUP BY OrderStatus;
```

	OrderStatus	Orders
1	Cancelled	250
2	Delivered	231
3	Pending	237
4	Returned	247
5	Shipped	235

Query 7: Top Products by Revenue

SQL Query:

```
SELECT Product,
SUM(TotalPrice) AS Revenue
FROM "Dataset for Data Analytics (2)"
GROUP BY Product
ORDER BY Revenue DESC
LIMIT 5;
```

Purpose:

To identify the top-performing products based on revenue.

Result:

SQL 1* ✕		
1	SELECT Product,	
2	SUM(TotalPrice) AS Revenue	
3	FROM "Dataset for Data Analytics (2)"	
4	GROUP BY Product	
5	ORDER BY Revenue DESC	
6	LIMIT 5;	

	Product	Revenue
1	Chair	195620.11
2	Printer	195612.61
3	Laptop	192126.56
4	Tablet	186568.95
5	Monitor	175651.41

5. KEY INSIGHTS

- SQL was successfully used to analyze the dataset.
- Total orders were calculated using COUNT().
- Total revenue was calculated using SUM().
- Average order value was calculated using AVG().
- Customer payment preferences were analyzed using GROUP BY.
- Order status distribution was examined.
- Top revenue-generating products were identified.

6. CONCLUSION

This project provided practical experience in SQL-based data analysis. Various SQL queries were used to retrieve, filter, group, and summarize data. The project enhanced understanding

of database querying techniques and demonstrated how SQL can be used to generate valuable business insights from raw datasets.